

External Communication and Project Performance: An Investigation into the Role of Gatekeepers

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ABSTRACT

This study investigates the role of certain boundary spanning individuals, labelled gatekeepers, in the transfer of information in an R&D setting by comparing the performance of project groups with and without gatekeepers. Results indicate that gatekeepers perform a linking role only for projects performing tasks that are locally oriented while universally oriented tasks were most effectively linked to external areas by direct project member communication. Evidence also suggests that gatekeepers do more than mediate external information; they may facilitate the external communication of their more local project colleagues.¹

INTRODUCTION

Oral communication networks represent an important medium through which information and new ideas are constantly transmitted both within and across organizational boundaries. Our research focuses on the effectiveness of the gatekeeping function within the actual communication networks of a major R&D facility. Gatekeepers are those key individual technologists who are both strongly connected to internal colleagues and strongly connected to sources of external technology. More specifically, we investigate the relationships between gatekeeper presence and project performance for different kinds of tasks and technical activities.

Gatekeepers and Technology Transfer

Organizational systems must collect and process information from outside sources in order to keep informed about relevant external developments and technological innovations. One way of dealing with the growing demands of information processing is through specialization -- certain specialized individuals evolving to keep abreast of relatively homogenous segments of the system's technological work environment. Such differentiation, in turn, is associated with the development of more locally defined languages and orientations; that is, a locally shared semantic and cognitive field to define, label, and generally organize a complex reality. Such localized definitions and specifications gradually unfold within projects from the different interactions among group members, the projects' overall objectives and task requirements, the common social and task related experiences of project members, and project norms, values, and historical perspectives (Price, 1965; Allen, 1977). These idiosyncratic developments are a basic determinant of attitudes and behaviors in that they strongly influence the ways in which project members think about and define their various problems and associated solution strategies.

This local orientation and coding scheme development is a double-edged sword. For those who share in this common language and awareness,

communication is remarkably efficient. On the other hand, if individuals do not share a common coding scheme and technical language, their work-related communication will be less efficient. This lack of perceptual and linguistic commonality can be conceptualized as communication impedance. The greater the mismatch in language and cognitive orientation, the greater the difficulties of communicating.

There is, then, a paradox. The evolution of more localized coding schemes helps projects deal with their more local information processing requirements; yet, it also hinders the acquisition and interpretation of information from external areas. Nevertheless, external information is vital in keeping abreast of the latest scientific and technological developments. How, then, can project groups be effectively linked to external information areas?

One way to deal with the difficulties of communicating across differentiated boundaries is through gatekeepers; individual project members who are strongly linked to sources of external technology and who are also capable of translating across contrasting coding schemes. With the help of these key individuals, external information can be channelled into the organization and its project groups by means of a two-step process (Allen and Cohen, 1969). First, gatekeepers are able to gather and understand external information, and subsequently, they are able to translate this information into terms that are more useful to their more locally constrained colleagues. Gatekeepers, as a result may be the principal means by which external technology can be effectively transferred into the organization.

While substantial literature supports the existence of gatekeepers, there is virtually no direct evidence to support the notion that gatekeepers can enhance project performance. Project SAPPHO (Achelladeles, Jervis, and Robertson, 1971) provided a case study, while Katz and Tushman (1979) and Allen, Tushman, and Lee (1979) provided only inferential support for the positive association between gatekeepers and project performance. The initial research question, then, investigates the association between gatekeepers and project performance. Furthermore, it is essential that we examine in more detail the role of gatekeepers with respect to information transfer. Must gatekeepers be the primary source of external information, or can they also serve to facilitate the external communication of their more locally oriented colleagues?

Gatekeepers and Project Performance

The need for a two-step process of information flow hinges on the existence of communication impedance between a project group and its external information areas. To the extent that different technical languages and coding schemes exist between project members and their external technological environments, communication across organizational boundaries will be difficult.

Several studies, for example, have found an inverse association between the amount of external communication and both individual and project performance.

One explanation for such negative relations comes from the idea that technological tasks (unlike the sciences) are strongly local in that their problems, strategies, and solutions are defined and operationalized in terms of the particular interests and orientations of the organizational subculture in which they are being addressed. As a result, project groups in different organizations may face similar problems yet may define their solution approaches and parameters very differently. And it is precisely because technological problems are typically defined in such local terms that most engineers have difficulty communicating effectively with outside professionals and consultants about their project-related activities. Locally oriented projects, therefore, will require gatekeepers to provide the necessary linkages to external areas, for direct contacts by other project members to sources of external technology are likely to be ineffective.

On the other hand, if external sources of information do not have different language and coding schemes from members of the project group, then it is less likely that this kind of communication impedance will exist. Work that is more universally defined (scientific work, for example) is probably less influenced and less constrained by local organizational factors. As a result, there is probably less of an impediment to external communication. Project members working on universal-type tasks, therefore, are simply more capable of understanding the nature of the problems and corresponding solution approaches employed by their relevant external colleagues. Scientists from one organization, for example, can easily communicate with scientists from any other organization about their overlapping sets of scientific interests. For universally defined tasks, then, gatekeepers may not be required to link projects with their relevant external information areas; instead, direct peer contact by all project members may be more advantageous.

The nature of a project's work, therefore, is a key factor affecting the development of localized languages and orientations and, as a result, may affect significantly the relationship between project performance and the presence of gatekeepers within project groups.

Role of Gatekeepers

As previously argued, gatekeepers are likely to facilitate the performance of project groups working on locally defined tasks. If this proposition is, in fact, supported empirically, then what are the contributions of gatekeepers such that their existence within a project seems to enhance its performance? One traditional explanation is that gatekeepers are a primary linking mechanism to external sources of information and technology; information flows through these key individuals to the more local members of the project team. From this perspective, relevant external information is transferred effectively into the project groups because of the capable boundary spanning activities

of the projects' gatekeepers.

A second explanation is that gatekeepers take an active training, development, and socialization role within their work groups. From this point of view, gatekeepers not only gather and channel external information, but they also facilitate the external communication of their project colleagues. Gatekeepers may work to reduce communication boundaries between their projects and external areas by directing, training, and coaching the external communications of their fellow project members. Under these conditions, both gatekeepers and other members of the project are able to effectively gather information from external areas.

If gatekeepers do permit other members to communicate effectively with external areas, then for projects with local tasks and gatekeepers, there should be a positive association between a project's external communication and its performance. If gatekeepers do not play this more active role, then there should be an inverse relation between a project's external communication and its performance.

METHODOLOGY

Research Setting

This study was carried out at the R&D facility of a large American Corporation employing a total of 345 professionals. The laboratory was organized into seven departments, each containing its own set of projects -- a total of 61 separate project groups existed across the seven departments. These projects remained stable over the data collection period, and each professional was a member of only one project.

Technical Communication

As discussed in Katz in Tushman (1979), each professional was asked to keep track of all other professionals with whom he or she had work-related oral communication on a particular day. This sociometric data was collected on a randomly chosen day each week for fifteen weeks. Respondents were asked to report all oral work related contacts both within and outside the laboratory's facility, including whom they talked to and how many times they talked with that person during the day.

Project communication is a measure of the average amount of technical communication per person per project over the fifteen weeks. Five mutually exclusive communication measures were operationalized for each project as follows: (1) communication within the Project; (2) communication within the project's Department (but not including project members); (3) communication to other Laboratory departments; (4) communication to other Organizational divisions (including marketing and manufacturing); (5) communication to external Professionals outside the organization (including consulting firms, universities, professional societies, suppliers, vendors, and customers). Individual responses were pooled to obtain measures of project communication with each of these various areas.

Conceptually, gatekeepers are defined as those internal stars (i.e., high internal communicators) who also maintain a high degree of external communication. (Allen, 1977). This study operationalized gatekeepers as those individuals

who were in the top fifth of their intra-department communication distribution and who were also in the top fifth of the extra-organizational communication distribution. Gatekeepers were identified in 20 project groups while 40 projects had no gatekeepers within their memberships.

Project Task Characteristics

In R&D settings, task can differ along several dimensions, including time span of feedback, specific vs. general problem-solving orientation, and generation of new knowledge vs. utilization of existing knowledge and experience (Rosenbloom and Wolek, 1970). Based on these dimensions, task categories were developed to form a universal (reserach) to local (technical service) task dimension. Four task categories were defined for (1) Basic Research; (2) Applied Research; (3) Development; and (4) Technical Service kinds of projects. (See Tushman, 1977 for specific definitions).

Using these definitions, respondents were asked to select the category which best characterized the objectives of their project and to indicate, on a three-point scale, how completely the project's objectives were represented by the selected category. The twelve possible answers were scored along a single scale ranging from completely basic research to completely technical service. Respondents were also asked to indicate what percentage of their project's work fell into each of the four categories. A weighted average of the percentages was calculated for each respondent. The scored responses to these two questions were averaged (Spearman-Brown reliability = .91).

Individual member responses were combined to get project scores. (Tushman, 1977). The distribution of project task scores clustered into three distinct categories: (1) Research (a combination of basic and applied research categories); (2) Development, and; (3) Technical Service. Research projects carried out universally oriented scientific work (for instance, developing new knowledge in glass physics), while development and technical service work was locally oriented in that they worked on organizationally defined problems and products.

Project Performance

A subjective measure similar to that used by Lawrence and Lorsch (1967) was employed to measure project performance. Each Department Manager (N=7) and laboratory director (N = 2) was separately interviewed and asked to evaluate the overall technical performance of all projects with which he was technically familiar. Based on these interviews, each project was independently rated by an average of 4.7 managers on a seven-point, Likert-type scale ranging from very low to very high. (Spearman-Brown reliability = .81).

Demographic Data

During the course of the study, demographic data was also collected from the laboratory's professionals, including their age, educational degrees, years in the laboratory, and years in their current project.

RESULTS

Gatekeepers and Project Performance

As previously discussed, there should be no overall main effect between the existence of gatekeepers and project performance. Instead, there should be significantly different relationships depending upon the projects' task characteristics.

As expected, the average performance of projects with gatekeepers (mean = 4.70) is not significantly different from the average performance of projects without gatekeepers (mean = 4.53). As a result, a two-way ANOVA was employed to test for the interaction effect between task characteristics and gatekeeper existence on project performance. As expected, there were no main effects on project performance for either the existence of gatekeepers or for task characteristics. There was, however, a strong interaction effect ($F = 4.73$; $P < .01$; $DF = 2, 54$).

More specifically, the breakdown of performance means strongly supports an inverse relation between gatekeeper presence and project performance. Research projects with gatekeepers were significantly lower performing than research projects without gatekeepers. In fact, the correlation between the existence of gatekeepers and project performance was significantly negative ($r = -.47$; $P < .05$). It may be that research projects are more effectively linked to external areas through direct member contact.

There is also partial support for a positive association between gatekeeper existence and project performance in that development projects with gatekeepers were significantly more effective than development projects without gatekeepers. In sharp contrast with research projects, the correlation between gatekeeper existence and development project performance was strongly positive ($r = .51$; $P < .01$). Unlike research projects, then, development projects seem to be linked to external areas more effectively through the use of gatekeepers. Technical service projects, on the other hand, exhibit no significant difference between projects with and without gatekeepers. As a result, the mechanisms used by technical service projects to import external information remain unclear.

Role of Gatekeepers

It was previously argued that for locally oriented tasks, gatekeepers may also act to reduce communication inpedance through the training, directing, and socializing of their fellow project colleagues. If gatekeepers serve this dual rôle then both gatekeepers and their peers will be able to communicate effectively with external information areas. In contrast, locally oriented projects without gatekeepers will have no clear effective link to external areas.

Results in the present study support these ideas. For development and technical service projects without gatekeepers there was a consistent inverse association between projects' extra-organizational communication and project performance. ($r = -.45^*$ and $r = -.19$ for development and technical service projects respectively; $*p < .05$). For locally oriented

projects with gatekeepers, however, a significantly different pattern emerged -- extra-organizational communication was positively associated with project performance. ($r = .31$ and $r = .32$ for development and technical service projects, respectively). Moreover, these positive correlations became even more significant after the direct communication effects of gatekeepers were removed! Finally, these significant correlational differences between local projects with and without gatekeepers strongly supports the argument that gatekeepers can have an important affect on the ability of project members to communicate effectively with external sources of technological information.

Alternative Explanations

Given the nature of the preceding results, alternative explanations were examined. For all pairwise correlational comparisons, means and standard deviations were checked to ensure that none were significantly different. Furthermore, it is important to make sure that the composition of projects with and without gatekeepers do not differ in some other meaningful way. It has been suggested that project performance and communication might be influenced by demographic characteristics including age, educational level, and project tenure, see Pelz and Andrews (1966) and Katz (1979). To rule out such rival possibilities, we compared the different project groupings along these important demographic variables. As there were no statistically significant differences, rival hypotheses based on such demographic differences are less plausible.

DISCUSSION

The awareness and acquisition of outside technology is vital for R&D based organizations. Moreover, there are at least two distinct methods by which R&D project groups can draw upon external technological developments: (1) direct contact by all project members; and (2) contact mediated by technological gatekeepers. The main purpose of our research was to compare the effectiveness of these alternative processes and secondly to examine more extensively the role that gatekeepers can play in facilitating the transfer of external information.

With respect to the first issue, our results suggest that the effectiveness of these two linking mechanisms is strongly contingent on the degree to which communication impedance separates project groups from their external information areas. Generally speaking, as project tasks become more specialized and more locally defined, it is likely that language and cognitive differences between the project and its extra-organizational domains will increase, intensifying communication impedance and resulting in more tendentious information flows. It is not that relevant and important information does not exist with outside sources, rather it simply becomes more difficult to mesh external ideas, suggestions, and solutions with internal technology that has become more locally defined and constrained. Therefore, for development projects, an inverse relation between external communication and project performance was

uncovered. Gatekeepers, as a result, became an especially important linking mechanism to external technology for these kinds of project groups.

With respect to more universally defined tasks, however, it was found that project members would be less constrained by local norms, values, and languages, resulting in less external communication impedance. For these research type tasks, it was found that direct interaction by all project members with their relevant external information areas was more effective than external contacts mediated through gatekeepers.

Contrary to expectations, however, the performances of technical service projects were not positively related to the presence of gatekeepers. One explanation for this contrasting result stems from differences in the nature of the work performed by development and technical service project groups. Development projects usually involve a more dynamic technology, new knowledge and/or new products. To keep abreast of such dynamic technologies and to introduce them effectively into appropriate development projects, what is needed are specialized individuals who keep current technically, are readily conversant with the technologies, and who are contributing to their project's work in direct and meaningful ways, i.e., technological gatekeepers (Allen, 1977).

Technical service project members, on the other hand, tend to work with more mature technologies, existing knowledge and/or existing products. Because these technologies are more stable and more easily dealt with by higher supervisory levels within the formal organization, the specialized gatekeeper role may not be needed to keep project colleagues informed about or to introduce new technological developments. Instead, the managerial hierarchy may be capable of keeping project members sufficiently informed about external events and information through formal operating channels.

Generally speaking, then, our findings seem to suggest that the beneficial effects of technical gatekeepers are strongly contingent on both the nature of the project's work and the stability of the involved technologies.

The Gatekeeping Role

What role do gatekeepers perform in linking local development projects to external areas? The data indicate that gatekeepers not only bring in technical information from external areas, but also facilitate the extra-organizational communication of their more locally oriented colleagues. In development projects, gatekeepers may actually increase the information processing capabilities of their groups by reducing the communication impedance separating most of their project colleagues from external areas. As a result, development projects with gatekeepers may be in a better position to take advantage of external technology since the number of members capable of communicating across organizational boundaries can increase, thereby, lessening their dependence on gatekeepers for gathering and disseminating external information. In research-type tasks, on the other hand, gatekeepers may not be a critical source of external information; nor does it appear that they

serve in any communication facilitating capacity. In higher performing research projects, members did not rely on specialized individuals for their external information. In a sense, they seemed to function as their own gatekeepers!

Conclusions

In conclusion, gatekeepers perform a critical role within R&D settings that often goes unrecognized. By realizing the importance of gatekeeping activities for development-type project tasks, R&D managers might be able to link their product and/or process development efforts to sources of external technology more effectively. A manager might want to examine, for example, the extent to which the more important technologies among the various development project activities are actually being "covered" by a gatekeeping type person. The degree to which gatekeeping activities can be managed, however, may be limited. Gatekeeping is an informal role in that other project engineers must feel sufficiently secure and comfortable psychologically to approach gatekeepers with their technical thoughts, problems, mistakes, and questions without fear of personal evaluation or other adverse considerations (Allen, 1977). Therefore, to the extent that the organization tries to formalize such a gatekeeping function, it runs the risk of inhibiting the very kinds of interaction that it wishes to promote.

This is not meant to imply that gatekeeping cannot be managed or helped; on the contrary, it can. In fact, a number of R&D facilities have instituted official gatekeeper programs. What is important to recognize is that the interest of gatekeepers in external technology cannot be suddenly "decreed" or assigned by management. Typically, such outside professional interests are a "given" and cannot be easily started or developed by the organization, although they can, of course, be made easier to pursue. What can be more easily influenced is the degree to which gatekeepers are actually present in project groups and used effectively in internal project interactions and activities. In short, the external side of the gatekeeping role is usually being performed by the gatekeeper anyway. It is the internal side that can be fostered, augmented, and made more effective.

FOOTNOTES

1. This is an equally co-authored piece of research, conducted at the R&D facility of a major midwest American Corporation. The authors would also like to thank Professor Thomas J. Allen for his collaborative efforts and ideas with respect to this research project. A more complete version of this paper can be found in Tushman and Katz (1980).
2. This research makes a basic distinction between gatekeepers and individuals who simply have substantial boundary spanning activities (BSA). To satisfy a gatekeeping function, an individual must be strongly connected both internally and externally. The assumption in many previous boundary spanning studies is that those individuals reporting high BSA are also

well-integrated internally, transferring and disseminating their information to others in the organization. Such an assumption, however, is often unjustified.

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